

Interface-driven electrocatalytic activity of NiFe-LDH on MnO₂-modified SWCNT hierarchical nanostructures for overall water splitting

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Abstract

Developing cost-effective bifunctional electrocatalysts is important for sustainable and clean energy production. Herein, a hierarchical NiFe layered double hydroxide integrated with MnO₂ nanorods and single-walled carbon nanotubes (NiFe-LDH@MnO₂/SWCNT) was synthesized to enhance bifunctional catalytic activity toward oxygen evolution reaction (OER) and hydrogen evolution reaction (HER). The synergistic coupling between Ni and Fe promotes high-valent Ni active sites that facilitate OH[−] adsorption and accelerate OER kinetics, whereas synergistic adsorption of hydrogen and hydroxyl intermediates facilitates improved HER performance. The incorporation of multivalent MnO₂ introduces redox-active centers that enhance electron transfer and intermediate formation, whereas the conductive SWCNT framework provides rapid charge transport and increased electrochemically active surface area. Among the synthesized catalysts, NiFe-LDH@3-MnO₂/SWCNT exhibits superior electrocatalytic performance, with low overpotentials of 206 mV for OER and 198 mV for HER at 10 mA cm^{−2} with a small OER Tafel slope of 33 mV dec^{−1}. Comparative analyses demonstrate that the catalyst outperforms several previously reported NiFe-LDH and MnO₂-based systems, exhibiting faster reaction kinetics and improved efficiency. The enhanced activity achieved from hierarchical architecture, improved electrolyte diffusion, and strong interfacial electronic coupling. These results reveal that NiFe-LDH@MnO₂/SWCNT is a promising electrocatalyst for efficient overall water splitting.

Keywords NiFe-LDH · MnO₂ · SWCNT · HER · OER · Water splitting

Introduction

Water electrolysis via electrocatalysis is emerging as a leading method for producing hydrogen (H₂) and oxygen (O₂) from water, offering a clean, sustainable pathway for energy production. As an environmentally friendly technology, it avoids greenhouse gas emissions and offers a viable alternative to fossil-fuel-based energy sources. Moreover, its scalability makes it attractive for industrial applications^{1,2}. The minimum theoretical cell voltage required for water splitting is 1.23 V, corresponding to a Gibbs free energy change (ΔG) of 237.1 kJ mol^{−1}. In practice, the required voltage exceeds this value due to sluggish kinetics of the hydrogen evolution reaction

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(HER) and the oxygen evolution reaction (OER), which significantly reduce energy conversion efficiency. Reducing the overpotential associated with these reactions is therefore critical, and the development of efficient electrocatalysts plays a central role in improving overall performance. While noble-metal-based catalysts, such as Pt (for HER) and Ru/Ir oxides (for OER), remain the benchmarks, their high cost and limited availability hinder their widespread deployment. As a result, considerable research has focused on discovering cost-effective and durable alternatives³. Among these, transition-metal-based materials, including phosphides⁴, sulfides⁵, oxides⁶, nitrides^{1,7}, and hydroxides⁸, have shown great potential for enhancing both HER and OER activities.

Layered double hydroxides (LDHs) have attracted considerable interest as bifunctional catalysts for both HER and OER⁹. In these two-dimensional materials, transition-metal cations occupy octahedral sites within MO_6 , while charge-compensating anions reside between stacked layers. LDHs composed of Fe, Co, and Ni are particularly effective under alkaline conditions: incorporation of trace Fe into $\text{Ni}(\text{OH})_2$ markedly improves OER activity by enhancing interfacial conductivity^{10,11}. Among them, NiFe-LDH stands out due to its unique electronic structure, large specific surface area, and tunable interlayer chemistry, which together confer excellent catalytic performance. Indeed, the identity of the intercalated anion can modulate the electron density at active sites and thus strongly influence OER kinetics^{12–14}. In particular, Reevesite-type NiFe-LDH, a naturally occurring hydroxide-phase mineral, showed enhanced OER activity compared to conventional NiFe-LDH. This improvement is attributed to potential-induced contraction of the M–O bond at Ni and Fe centers, which optimizes the adsorption energies of oxygen intermediates and lowers overpotentials. Moreover, the strong electronic interactions between Ni and Fe facilitate efficient redox transitions and accelerate charge-transfer processes¹⁵.

The intrinsic activity of NiFe-LDH can be further boosted through hybridization with conductive carbon nanomaterials such as carbon nanotubes, graphitic carbon nitride, and graphene^{16,17}. These carbonaceous scaffolds (i) provide rapid electron pathways, (ii) supply additional catalytic sites, (iii) suppress nanosheet agglomeration, and (iv) facilitate mass transport of reactants and products^{18,19}. For example, coupling NiFe-LDH with graphene oxide yields a highly percolated network that accelerates both charge transfer and mass diffusion²⁰. Likewise, single-walled carbon nanotube (SWCNT)– In_2O_3 heterostructures promote electron transfer through the metallic nanotube backbone while In_2O_3 enhances the adsorption of water and reaction intermediates²¹. Uniform wrapping of MnO_2 around multiwalled CNTs has been shown to increase catalytic activity dramatically, and SWCNTs themselves afford high conductivity, a large surface area, and favorable hydrogen-adsorption energetics^{22–24}. MnO_2 is an earth-abundant material, making it an attractive co-catalyst. Mixed-valent $\text{Mn}^{4+}/\text{Mn}^{3+}$ centres facilitate rapid redox cycling, significantly enhancing the electrocatalytic performance of OER and HER. Furthermore, the unique crystallographic structure of MnO_2 , arising from the arrangement of MnO_6 octahedra, improves electrocatalytic efficiency^{25,26}. These structural features translate into low charge-transfer resistance and strong O_2 adsorption, resulting in superior OER performance for MnO_2 relative to many other first-row oxides MnO_2 compared with many conventional first-row transition-metal oxides^{27,28}. Incorporating Mn into NiFe-LDH has been shown to modulate its electronic structure and catalytic behavior further. Mn doping introduces hybrid energy levels between the Ni 3d and O 2p orbitals, effectively narrowing the band gap and improving electrical conductivity. This modification also reduces the Gibbs free energy required for the adsorption and conversion of oxygen intermediates, thereby facilitating the $\text{Ni}^{2+} \rightarrow \text{Ni}^{3+}/\text{Ni}^{4+}$ transition, which is important for efficient OER kinetics. Moreover, Mn incorporation optimizes the electron density distribution around Ni and Fe centers, enhancing their ability to adsorb and activate reaction intermediates and thereby improving the overall catalytic efficiency of NiFe-LDH-based catalysts^{29,30}.

Despite extensive studies on NiFe-LDH-based electrocatalysts, achieving low overpotentials and rapid OER kinetics remains a key challenge. Previously reported systems such as $\text{Ni}_2\text{Fe}_1\text{-LDH/rGO/NF}$ (228 mV, 37.9 mV dec^{-1})³¹, $\text{NiMoO}_4/\text{NiFe-LDH/rGO}$ (270 mV, 76.2 mV dec^{-1})³², and $\text{NiFe-LDH/Ti}_3\text{C}_2\text{T}_x\text{-rGO}$ (~ 235 mV, ~ 40 mV dec^{-1})³³ highlight the need for further enhancement in catalytic efficiency. The present NiFe-LDH integrated with SWCNT and MnO_2 exhibited a lower Tafel slope (~ 33 mV dec^{-1}) and reduced overpotential, indicating faster kinetics and improved charge-transfer efficiency. This enhanced performance is attributed to the synergistic

interaction between the conductive SWCNT network and redox-active MnO_2 , which promotes efficient electron transport and increases active site accessibility. Notably, the exploration of the Reevesite phase combined with such a ternary architecture provides a unique strategy that remains largely unexplored for overall water splitting applications.

The remainder of this article is organised as follows. Section 2 presents the synthesis procedure; Sect. 3 discusses the structural, morphological, and electrochemical analyses of the composite catalyst; and Sect. 4 summarizes the key findings and highlights the broader significance of this work.

Experimental section

Materials and chemicals

All chemicals used, such as ferric nitrate nonahydrate, nickel (II) nitrate hexahydrate, and urea, were obtained from SRL, India. Super black and polyvinylidene fluoride (PVDF) were sourced from Sigma-Aldrich, while single-walled carbon nanotubes (SWCNTs) and potassium permanganate were supplied by Nano Research Elements and Merck, respectively. Deionized water purified through a Milli-Q system was used in all procedures. All chemicals were of high purity and used without further purification, with specifications as provided by the respective suppliers.

Synthesis of NiFe-LDH

A solution was prepared by dissolving 1 mmol of nickel nitrate hexahydrate and 0.5 mmol of ferric nitrate nonahydrate in a mixture of 50 mL of water and 30 mL of ethanol. To this, 4 mmol of urea was introduced, and the mixture was stirred until fully mixed. The homogeneous solution was poured into a stainless steel autoclave lined with Teflon and subjected to hydrothermal treatment at 120 °C for 16 h. The resulting solid was separated by filtration, thoroughly washed with both deionized water and ethanol, and then dried under vacuum at 80 °C.

Synthesis of MnO_2 /SWCNT

A dispersion of 20 mg of SWCNTs was prepared in 100 mL of deionized water via ultrasonication. To this homogeneous suspension, 200 mg of KMnO_4 was added, and the mixture was stirred continuously for 1 h. Subsequently, concentrated hydrochloric acid was slowly introduced to reduce the permanganate ions, and stirring was continued for an additional hour. The resulting mixture was transferred to a Teflon-lined stainless steel autoclave and subjected to hydrothermal treatment at 140 °C for 1 h. After naturally cooling to room temperature, the product was washed thoroughly with deionized water and ethanol to remove residual reactants and impurities, followed by drying at 70 °C.

Synthesis of NiFe-LDH@ MnO_2 /SWCNT

MnO_2 /SWCNT (1, 3, 5 mg) was dispersed in a 1:1 mixture of DI water and ethanol for 1 h, and then the NiFe-LDH synthesis procedure was followed for the synthesis of NiFe-LDH@ MnO_2 /SWCNT. The 1, 3, and 5 mg MnO_2 /SWCNT samples with NiFe-LDH are named NiFe-LDH@1- MnO_2 /SWCNT, NiFe-LDH@3- MnO_2 /SWCNT, and NiFe-LDH@5- MnO_2 /SWCNT, respectively. The schematic representation of the synthesis procedure is given in Fig. 1.

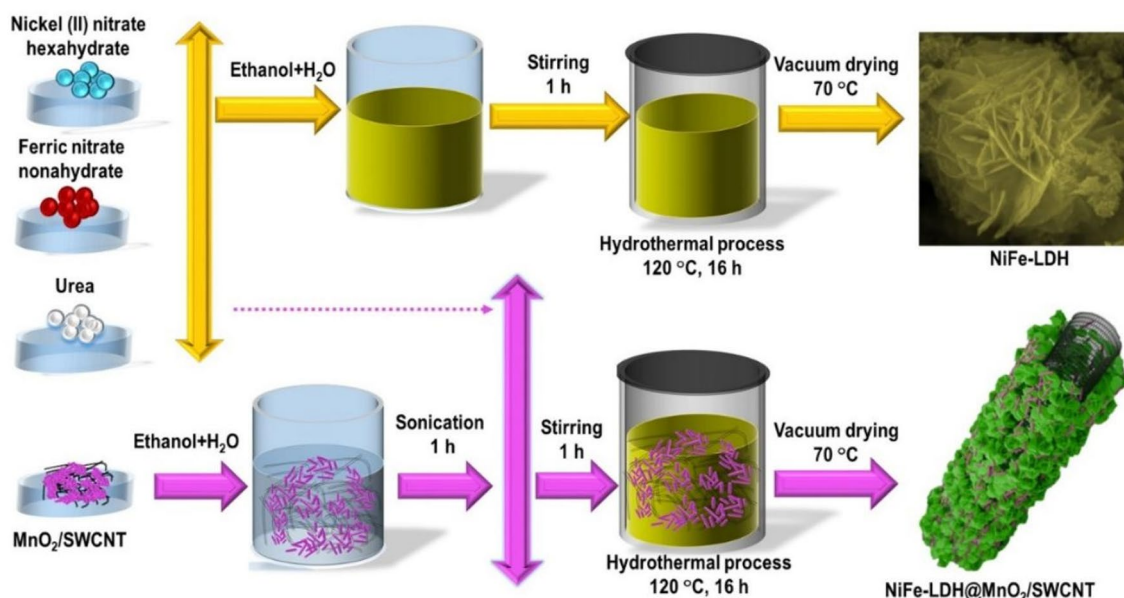


Fig. 1 Schematic representation of the synthesis procedure of NiFe-LDH and NiFe-LDH@MnO₂/SWCNT.

Material characterization

The samples were characterized using various techniques, including XRD on a PANalytical EMPYREAN instrument, SEM on a Carl Zeiss EVO 10 microscope, HR-TEM on a JEOL JEM-2100 Plus microscope, and XPS on a ThermoScientific K-Alpha instrument.

Electrochemical measurements

The electrochemical tests were performed using a Squidstat™ Plus electrochemical workstation in a three-electrode setup with Hg/HgO reference electrode and graphite rod counter electrode. The graphite rod counter electrode was used to avoid possible interference from Pt-based counter electrodes during HER measurements, while Hg/HgO was selected as a pH-compatible reference electrode for alkaline electrolyte, providing a stable reference potential in 1.0 M KOH^{34,35}. The working electrode was prepared by coating a nickel foam piece (1 cm x 1 cm) with a catalyst slurry, comprising the catalyst, carbon black, and PVDF in a 9:0.5:0.5 mass ratio, and drying it in a vacuum oven at 80 °C for 12 h. The deposited coating mass was approximately 3.5 mg on a 1 cm x 1 cm electrode, corresponding to a total loading of $\sim 3.5 \text{ mg cm}^{-2}$. Carbon black was used as a conductive additive to improve electron transport within the catalyst layer, while PVDF was used as a binder to enhance the adhesion of the catalyst on the nickel foam substrate. Control experiments showed that direct coating of the catalyst on nickel foam without binder led to poor adhesion and delamination during electrochemical testing. For comparison, bare nickel foam and PVDF/CB-coated nickel foam were also evaluated as control electrodes under identical conditions, and the corresponding LSV curves are provided in Fig. S1 and Fig. S2. The electrochemical tests were conducted in 1.0 M KOH solution, involving linear sweep voltammetry (LSV) at a scan rate of 5 mV s^{-1} and electrochemical impedance spectroscopy (EIS). The results were referenced to the reversible hydrogen electrode (RHE) using the equation: $E_{\text{RHE}} = E_{\text{Hg/HgO}} + 0.059 \text{ pH} + 0.098$. Both the LSV curves and chronopotentiometric stability measurements were corrected for solution resistance using *i*R compensation.

Results and discussion

Structural and morphological analysis of NiFe-LDH@MnO₂/SWCNT nanocomposite

XRD analysis revealed the crystal structure of the electrocatalysts, as shown in Fig. 2a and b. The diffraction peaks at 11.7°, 19.5°, 34.4°, 39°, and 46.6° for NiFe-LDH and NiFe-LDH@MnO₂/SWCNT correspond to the Reevesite crystal phase, indicating the formation of a (Ni, Fe) hydrotalcite structure with carbonate anions in the interlayer space. This confirms the formation of NiFe-LDH with carbonate interlayer anions³⁶. The observed low-angle reflection at 11.7° corresponds to the basal reflection of the layered hydroxide structure. While the relative peak intensities differ slightly from the standard reference pattern (JCPDS No. 26-1286), such variations are attributed to stacking disorder and imperfect layer alignment, which are commonly observed in layered hydroxide materials³⁷. Furthermore, the successful in-situ growth of MnO₂ on SWCNT in MnO₂/SWCNT was verified by the appearance of MnO₂ peaks at 22.3°, 34.7°, 37°, 43°, 56.7°, and 62.5° (JCPDS No. 14-0644) in Fig. 2b. The peaks corresponding to MnO₂ are not distinguishable in the NiFe-LDH@MnO₂/SWCNT composite, likely due to its low concentration. Therefore, the presence of MnO₂ was confirmed by further characterization.

The morphology of the synthesized materials was studied using HR-TEM and SEM analysis. As shown in Fig. 3a, the synthesized NiFe-LDH@3-MnO₂/SWCNT exhibits a flower-like architecture comprising ultrathin nanosheets with expanded interlayer spacing. Notably, Fig. 3a (inset) also reveals separated sheets, indicating the presence of both stacked and individual nanosheets. The hierarchical nanosheet structures were developed via a two-step process: an initial oxidation reaction followed by crystallization. The nanosheets are stacked into a flower-like structure to reduce surface energy. The hierarchical, flower-like morphology of NiFe-LDH provides a greater number of surface-active sites, thereby enhancing adsorption and electrochemical performance³⁸. When NiFe-LDH is grown in situ on MnO₂/SWCNT, the hydrolysis of urea releases OH[−] ions that react with Ni²⁺ and Fe³⁺ ions to nucleate into nanosheets on the MnO₂/SWCNT surface. As the reaction progresses, these nanosheets stack and form a flower-like structure. The SWCNT serves as a conducting network that connects nanosheets and microflowers. MnO₂ nanorods are evenly distributed between the layers, as inferred from Fig. 3b³⁹. The micrographs of NiFe-LDH@3-MnO₂/SWCNT composite further confirmed the sheet-like morphology of NiFe-LDH. The images reveal the formation of defect-rich hexagonal nanosheets that interconnect and stack upon each other. The characteristic hexagonal shape of LDH is evident in Fig. 3c and d, where MnO₂ and SWCNT are embedded between the NiFe-LDH sheets. Figure 3e shows the particle size distribution curve, indicating an average MnO₂

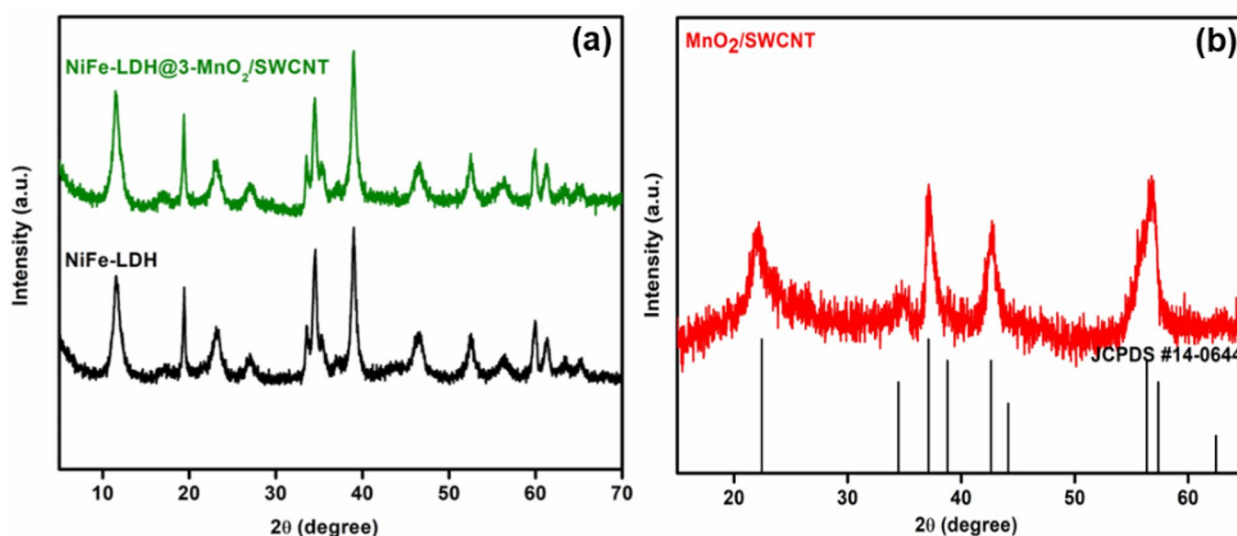


Fig. 2 XRD spectrum of (a) NiFe-LDH, NiFe-LDH@MnO₂/SWCNT, and (b) MnO₂/SWCNT.

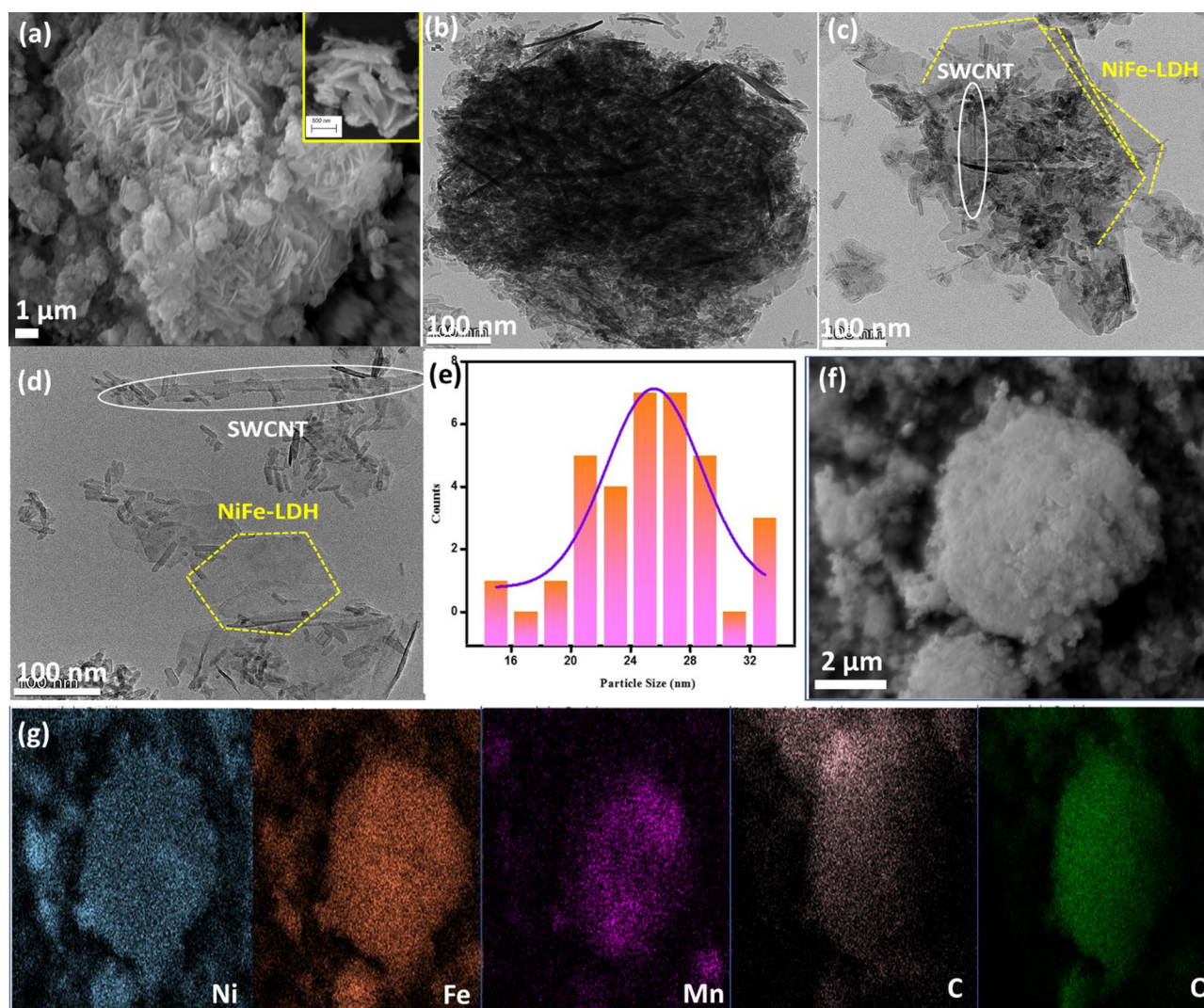


Fig. 3 (a) SEM image of flower-like NiFe-LDH@3-MnO₂/SWCNT (inset: NiFe-LDH nanosheet), (b–d) HR-TEM images of NiFe-LDH@3-MnO₂/SWCNT, (e) Particle size distribution curve of MnO₂, (f) SEM image of NiFe-LDH@3-MnO₂/SWCNT, and (g) its corresponding elemental mapping for Ni, Fe, Mn, C, and O.

nanorod size of approximately 25 nm. The MnO₂ nanorod decoration on NiFe-LDH nanosheets, combined with interconnected SWCNTs, enhances electrocatalytic activity and electrical conductivity by increasing the number of active sites. The elemental distribution in NiFe-LDH@3-MnO₂/SWCNT was confirmed using SEM color mapping (Fig. 3f), which revealed a uniform distribution of Ni, Fe, Mn, C, and O (Fig. 3g).

The XPS analysis confirmed the electronic state and electron migration in NiFe-LDH and NiFe-LDH@3-MnO₂/SWCNT. The XPS peak of MnO₂/SWCNT-loaded NiFe-LDH showed a negative shift in the Ni and Fe core level binding energies compared to pure NiFe-LDH. The XPS spectra were analyzed using a Shirley-type background correction, and the peaks were fitted with Gaussian profiles to precisely extract binding energies and identify the corresponding chemical states. The XPS analysis of NiFe-LDH (Fig. 4a) reveals the presence of nickel in the + 2 oxidation state, as evidenced by the Ni 2p_{3/2} and Ni 2p_{1/2} peaks at 856.8 and 874.4 eV, respectively, accompanied by satellite peaks at 862.5 and 880.3 eV⁴⁰. These peaks were shifted to lower binding energies when NiFe-LDH was modified with MnO₂/SWCNT. The negative shift confirmed an increase in electron density around Ni due to the electron transfer by the MnO₂/SWCNT. Similarly, the Fe 2p spectrum of NiFe-LDH showed a negative shift on comparing NiFe-LDH@3-MnO₂/SWCNT. The Fe 2p spectrum of NiFe-LDH in Fig.

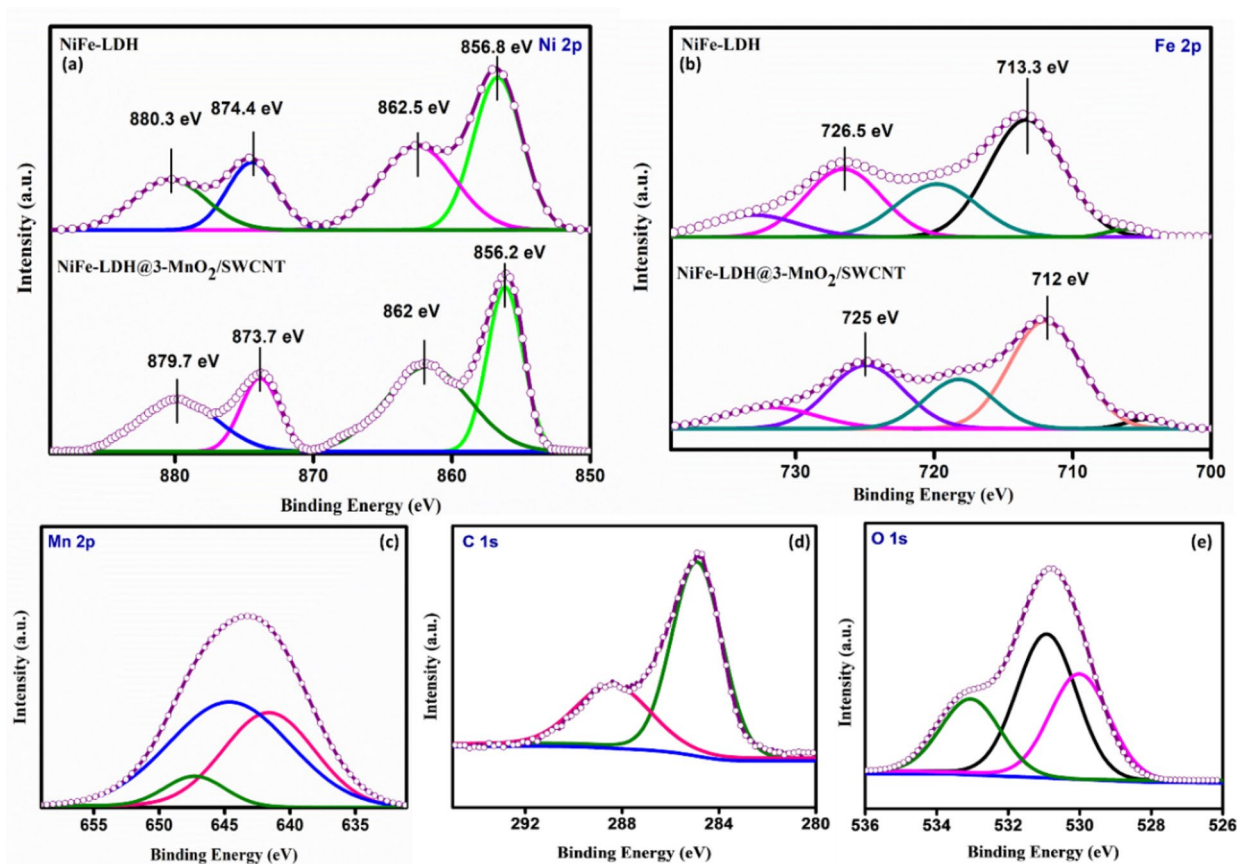


Fig. 4 XPS analysis of NiFe-LDH@3-MnO₂/SWCNT. The high-resolution spectrum of (a) Ni 2p (b) Fe 2p, (c) Mn 2p, (d) C 1s, and (e) O 1s.

4b confirmed the existence of Fe³⁺. The Fe 2p spectrum shows peaks at 713.3 eV and 726.5 eV, corresponding to the Fe 2p_{3/2} and Fe 2p_{1/2}, respectively. Satellite peaks at 719.8 eV and 733.04 eV confirm the presence of iron in the +3 oxidation state (Fe³⁺) in the catalyst. Electron transfer from MnO₂/SWCNT to NiFe-LDH facilitates the electrocatalytic reaction by enhancing the adsorption and desorption of reaction intermediates. The negative shift also confirmed the hetero-interface formed between NiFe-LDH, MnO₂, and SWCNT^{41,42}. The Mn⁴⁺ and Mn³⁺ peaks are observed at 644.60 and 641.6 eV, respectively, in Fig. 4c. A satellite peak is also present at ~647 eV, further confirming the presence of Mn in a mixed-valence state. In Fig. 4d, the peaks centred at 284.8 and 288.4 eV in the C 1s spectrum pertain to C-O in carbonate ions and O = C-O, respectively. The strong peak at 531 eV in O 1s (Fig. 4e) pertains to oxygen in the carbonate group, and the peaks at 533 and 529.9 eV correspond to M-H and M-O, respectively. The XPS results confirmed the formation of LDH with CO₃²⁻ intercalation and the presence of MnO₂ in the NiFe-LDH@3-MnO₂/SWCNT composite^{43,44}.

Electrochemical OER and HER performance of NiFe-LDH nanocomposites

The OER performance of NiFe-LDH, NiFe-LDH@1-MnO₂/SWCNT, NiFe-LDH@3-MnO₂/SWCNT, and NiFe-LDH@5-MnO₂/SWCNT was evaluated and shown in Fig. 5a. Overpotentials at 10 mA cm⁻² were calculated from LSV curves (Fig. 5a) as 223, 211, 206, and 221 mV for the respective catalysts. The improved OER performance of NiFe-LDH@3-MnO₂/SWCNT can be attributed to the synergistic combination of NiFe-LDH, SWCNT, and MnO₂. The OER control LSV curves (Fig. S1) show that NiFe-LDH/SWCNT exhibits better activity than SWCNT/MnO₂, indicating that NiFe-LDH provides OER-active Ni/Fe sites, while the SWCNT network improves

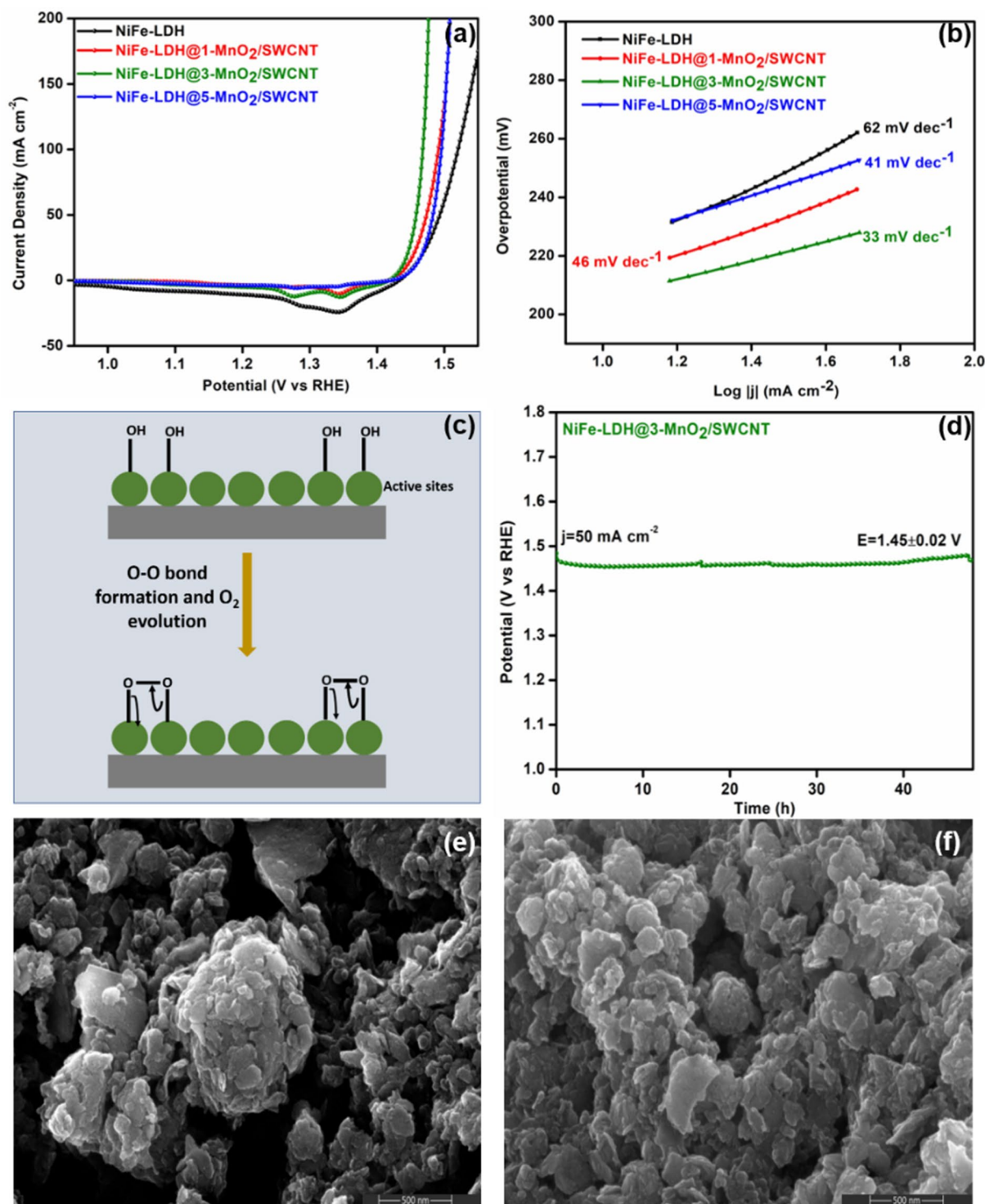


Fig. 5 (a) OER-LSV polarization, (b) Tafel plots of NiFe-LDH, NiFe-LDH@1-MnO₂/SWCNT, NiFe-LDH@3-MnO₂/SWCNT, and NiFe-LDH@5-MnO₂/SWCNT, (c) Mechanistic pathway of closely arranged active sites, (d) Chronopotentiometric analysis of NiFe-LDH@3-MnO₂/SWCNT at a constant current density of 50 mA cm^{-2} for 48 h, and (e, f) SEM images of NiFe-LDH@3-MnO₂/SWCNT after OER stability analysis.

electrical conductivity and charge-transfer pathways. However, the activity of these binary control electrodes is still lower than that of the optimized NiFe-LDH@3-MnO₂/SWCNT catalyst, confirming that the superior performance arises from the combined synergistic interaction of NiFe-LDH active sites, MnO₂ redox-active interfaces, and the conductive SWCNT network. The unique hybrid morphology of NiFe-LDH, featuring nanosheets and microflowers, facilitates enhanced electrolyte diffusion and improved electron conductivity. Moreover, the interconnected MnO₂ nanoparticles within the LDH nanosheets act as a co-catalyst, boosting water-oxidation efficiency. The incorporation of SWCNT further enhances conductivity and stability by forming a network that interconnects the LDH and MnO₂³⁹.

The Tafel slopes (Fig. 5b) were evaluated to investigate the electrocatalytic reaction kinetics using the equation $\eta = b \log j + a$, where η is the overpotential, b is the Tafel slope, j is the current density, and a is the Tafel constant. This analysis provides insight into the rate-determining step and efficiency of the catalyst. The Tafel slope was evaluated following the standard recommendations for electrocatalytic water-splitting analysis reported by Anantharaj et al.⁴⁵. The NiFe-LDH@3-MnO₂/SWCNT (33 mV dec⁻¹) possessed a smaller Tafel slope when compared to other catalysts NiFe-LDH (62 mV dec⁻¹), NiFe-LDH@1-MnO₂/SWCNT (46 mV dec⁻¹), and NiFe-LDH@5-MnO₂/SWCNT (41 mV dec⁻¹). This implies rapid electron transfer and intrinsic activity of the hybrid structure in NiFe-LDH@3-MnO₂/SWCNT. The observed Tafel slope of less than 40 mV dec⁻¹ suggests an efficient four-electron transfer process. Additionally, the oxidation of hydroxide species, oxygen-oxygen bond formation, and oxygen evolution co-occur. The electrocatalyst with closely spaced active sites will exhibit a similar mechanism, as shown in Fig. 5c⁴⁶. The chronopotentiometry technique (Fig. 5d) was applied to assess the stability of NiFe-LDH@3-MnO₂/SWCNT. The NiFe-LDH@3-MnO₂/SWCNT was tested for 48 h at a constant current density of 50 mA cm⁻². The NiFe-LDH@3-MnO₂/SWCNT showed excellent stability of about 98.04%. The observed post-OER morphology in Fig. 5e and f indicates electrochemical reconstruction of NiFe-LDH@3-MnO₂/SWCNT. This morphological changes can be attributed to OER-induced surface reconstruction and structural degradation of the NiFe-LDH-based composite under alkaline medium. During OER, the surface and edge sites of NiFe-LDH actively participate in the reaction, and continuous interaction with OH⁻ species can induce structural changes within the layered framework, leading to partial dissolution and reconstruction of the catalyst during prolonged OER operation⁴⁷.

The HER performance is also high for NiFe-LDH@3-MnO₂/SWCNT when compared to NiFe-LDH, NiFe-LDH@1-MnO₂/SWCNT, and NiFe-LDH@5-MnO₂/SWCNT, as shown in Fig. 6a. From the result, it is evident that the hetero-interface formed between NiFe-LDH, MnO₂, and SWCNT enhanced the HER performance of the modified catalysts. The synergistic contribution of these components to HER activity is further confirmed in Fig. S2. Specifically, NiFe-LDH@3-MnO₂/SWCNT exhibits the highest activity, achieving 10 mA cm⁻² at an overpotential of 198 mV. The overpotentials required for NiFe-LDH, NiFe-LDH@1-MnO₂/SWCNT, and NiFe-LDH@5-MnO₂/SWCNT to reach the current density of 10 mA cm⁻² are 248, 225, and 216 mV, respectively. Tafel values (Fig. 6b) for NiFe-LDH (127 mV dec⁻¹), NiFe-LDH@1-MnO₂/SWCNT (121 mV dec⁻¹), NiFe-LDH@3-MnO₂/SWCNT (119 mV dec⁻¹), NiFe-LDH@5-MnO₂/SWCNT (121 mV dec⁻¹), indicate that NiFe-LDH@3-MnO₂/SWCNT follows the Volmer-Heyrovsky mechanism in alkaline medium. Chronopotentiometry at 50 mA cm⁻² for 48 h (Fig. 6c) was used to assess the stability of NiFe-LDH@3-MnO₂/SWCNT. The NiFe-LDH@3-MnO₂/SWCNT exhibited remarkable stability of approximately 98%. The SEM image (Fig. 6d) of NiFe-LDH@3-MnO₂/SWCNT after the HER stability test shows that the catalyst maintains morphology on the electrode surface. Although slight surface roughening and particle aggregation are observed after prolonged HER operation, the overall structure remains well retained without severe collapse or detachment. This indicates good morphological stability of the catalyst during HER in alkaline electrolyte.

To investigate the role of electrochemically active sites in enhancing the activity of NiFe-LDH@3-MnO₂/SWCNT compared to NiFe-LDH, cyclic voltammetry measurements were performed at various scan rates (10–50 mV/s) in the non-faradaic region. The double-layer capacitance (C_{dl}) values for NiFe-LDH (3.33 mF/cm²), NiFe-LDH@1-MnO₂/SWCNT (3.77 mF/cm²), NiFe-LDH@3-MnO₂/SWCNT (4.79 mF/cm²), and NiFe-LDH@5-MnO₂/SWCNT (4.06 mF/cm²) were calculated from the data in Fig. 7(a-d), respectively, using the

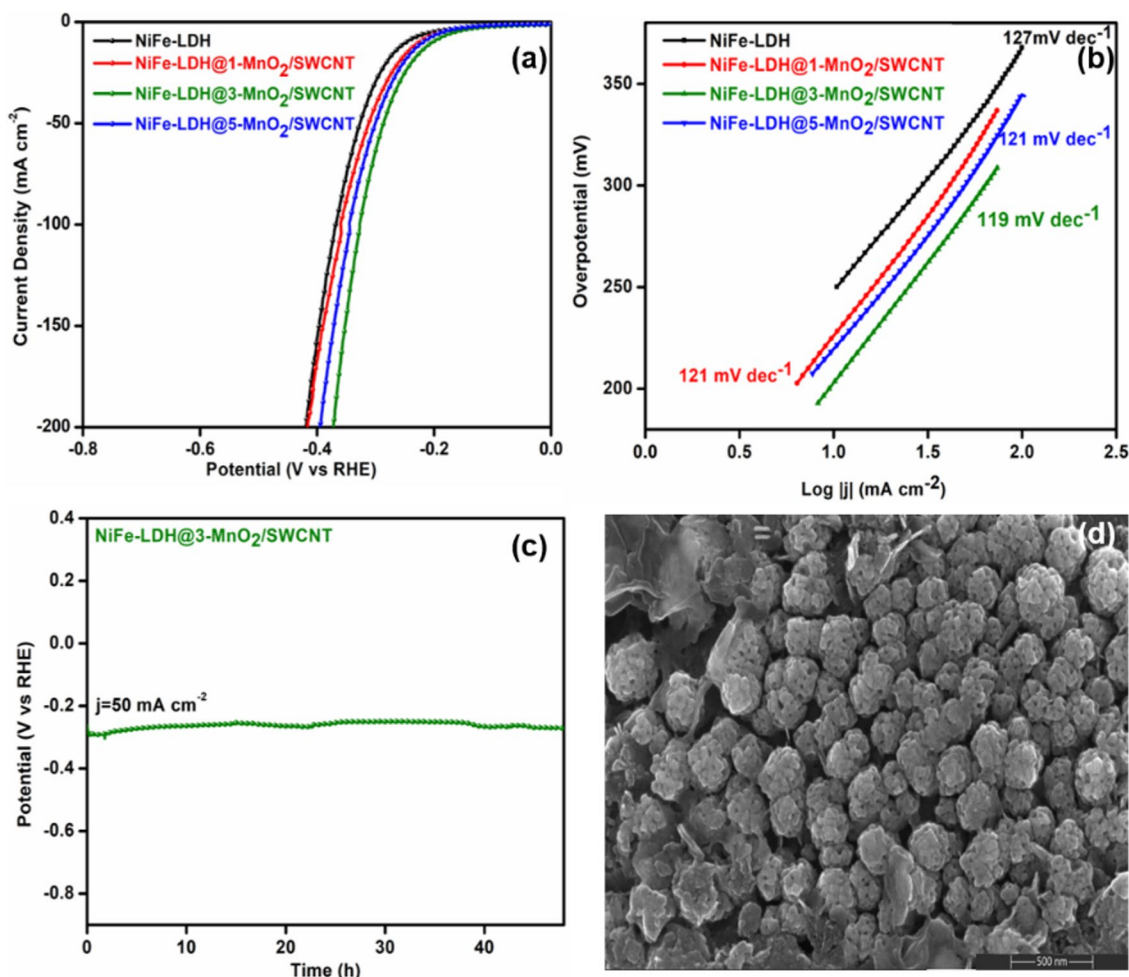


Fig. 6 (a) HER-LSV polarization and (b) Tafel plots of NiFe-LDH, NiFe-LDH@1-MnO₂/SWCNT, NiFe-LDH@3-MnO₂/SWCNT, and NiFe-LDH@5-MnO₂/SWCNT, (c) Chronopotentiometric analysis of NiFe-LDH@3-MnO₂/SWCNT at a constant current density of 50 mA cm⁻² for 48 h, and (d) SEM images of NiFe-LDH@3-MnO₂/SWCNT after HER stability analysis.

slope of Δj ($j_{\text{anodic}} - j_{\text{cathodic}}$) vs. scan rate. The electrochemical surface area (ECSA) was estimated by dividing the double-layer capacitance (C_{dl}) by the specific capacitance ($C_s = 0.040 \text{ mF cm}^{-2}$) and multiplying by the geometric area of the electrode (1 cm^2), providing insight into the catalyst's active surface area. The ECSA of the NiFe-LDH@3-MnO₂/SWCNT (119.75 cm^2) is higher than that of NiFe-LDH@5-MnO₂/SWCNT (101.50 cm^2), NiFe-LDH@1-MnO₂/SWCNT (94.3 cm^2), and NiFe-LDH (83.25 cm^2). ECSA normalization was performed to distinguish the intrinsic catalytic activity of the electrode from the activity arising from electrochemically active surface area. The ECSA normalized LSV plots for OER and HER in Fig. S3 were obtained by first normalizing the measured current with respect to the exposed geometric electrode area to obtain the geometric current density. Since the exposed geometric area of the electrode was 1 cm^2 , the measured current was divided by 1 cm^2 . The obtained geometric current density was then divided by the calculated ECSA to determine the ECSA-normalized current density. The ECSA-normalized HER and OER results confirm that the superior bifunctional electrocatalytic performance of NiFe-LDH@3-MnO₂/SWCNT arises from both its larger electrochemically active surface area and its improved intrinsic active-site activity⁴⁸. This enhancement is attributed to the hetero-architecture of NiFe-LDH@3-MnO₂/SWCNT, where the MnO₂ and SWCNT on NiFe-LDH provide a highly conductive pathway for efficient charge and ion diffusion. Moreover, the strong interaction among MnO₂, SWCNT, and

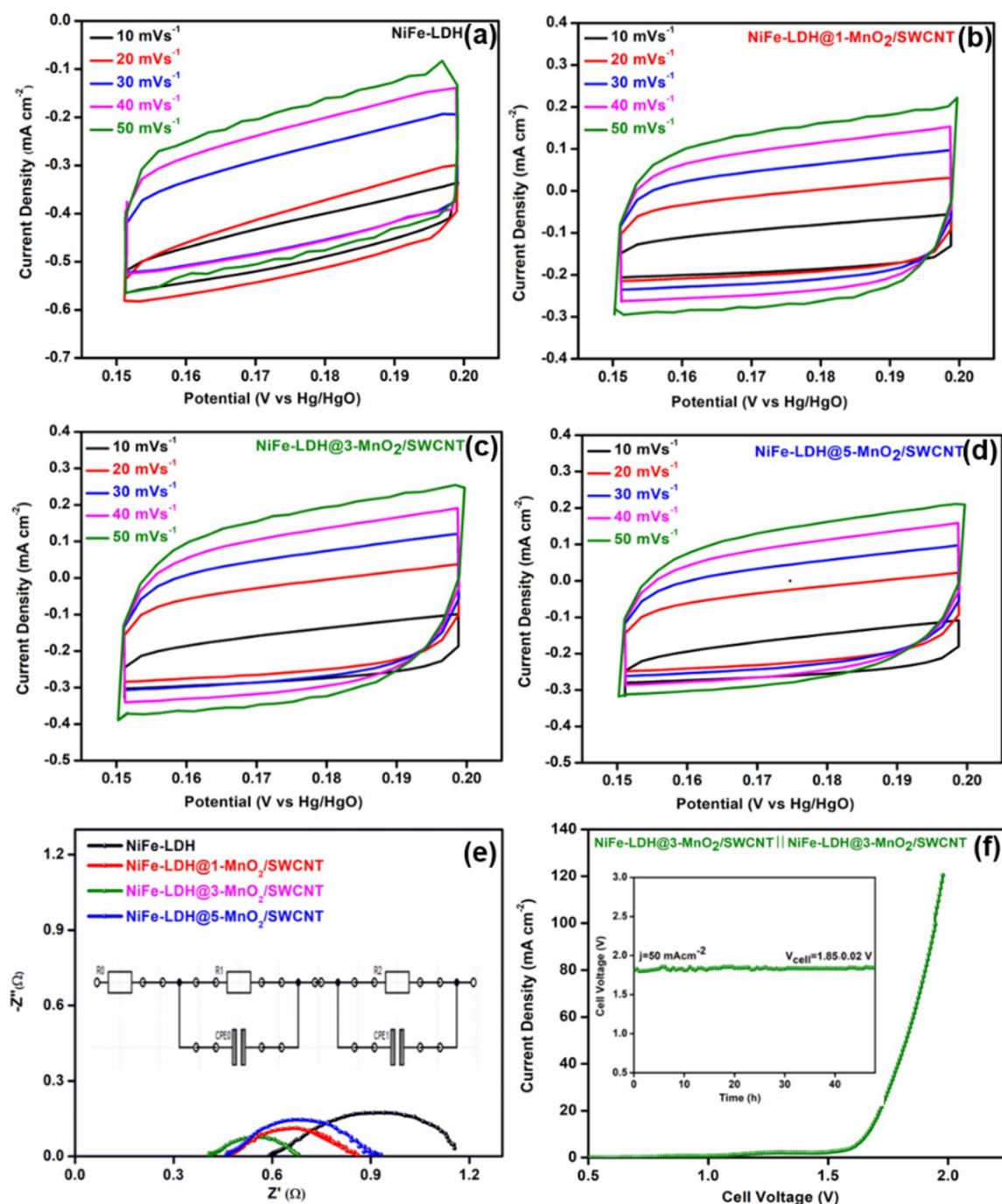


Fig. 7 CV curves of (a) NiFe-LDH, (b) NiFe-LDH@1-MnO₂/SWCNT, (c) NiFe-LDH@3-MnO₂/SWCNT, and (d) NiFe-LDH@5-MnO₂/SWCNT at the scan rates from 10 mV to 50 mV s⁻¹, (e) EIS plot of NiFe-LDH and its composites, and (f) LSV curve of NiFe-LDH@3-MnO₂/SWCNT||NiFe-LDH@3-MnO₂/SWCNT (inset: Chronopotentiometric curve of NiFe-LDH@3-MnO₂/SWCNT at constant current density of 50 mA cm⁻²).

NiFe-LDH provides more active sites and accelerates reaction kinetics. These results confirmed that the NiFe-LDH@3-MnO₂/SWCNT hetero-architecture effectively enhanced the overall electrocatalytic performance.

Furthermore, the lower solution resistance (R_s) and charge transfer resistance (R_{ct}) of NiFe-LDH@3-MnO₂/SWCNT (0.399 Ω; 0.265 Ω) than NiFe-LDH@5-MnO₂/SWCNT (0.450 Ω; 0.450 Ω), NiFe-LDH@1-MnO₂/

SWCNT (0.461 Ω ; 0.366 Ω), and NiFe-LDH (0.598 Ω ; 0.570 Ω) in Fig. 7e suggest enhanced electrocatalytic kinetics compared to NiFe-LDH. From the results, the optimized NiFe-LDH@3-MnO₂/SWCNT showed better performance than the others due to efficient charge transport and increased utilization of active sites. Having exhibited remarkable OER and HER activities, NiFe-LDH@3-MnO₂/SWCNT was subsequently employed as a bifunctional catalyst for overall water splitting. The overall water-splitting performance was evaluated in a two-electrode beaker-type cell containing 1.0 M KOH electrolyte, where NiFe-LDH@3-MnO₂/SWCNT-coated electrodes were used as both the anode and cathode. The corresponding full-cell LSV polarization curve of NiFe-LDH@3-MnO₂/SWCNT||NiFe-LDH@3-MnO₂/SWCNT is shown in Fig. 7f. The NiFe-LDH@3-MnO₂/SWCNT||NiFe-LDH@3-MnO₂/SWCNT system achieved cell voltage (V_{cell}) of 1.66 V at 10 mA cm⁻². The stability of the full cell was evaluated via chronopotentiometry at 50 mA cm⁻² for 48 h in 1.0 M KOH electrolyte in beaker-type cell. The NiFe-LDH@3-MnO₂/SWCNT||NiFe-LDH@3-MnO₂/SWCNT showed exceptional stability of about 98.63% after 48 h of continuous reaction.

Comparison of synthesized catalyst

Table 1 presents the electrocatalytic activity of the synthesized electrocatalysts. The synthesized NiFe-LDH composites have shown excellent OER performance and reasonable HER performance. The 3d transition metal hydroxide is more active in an alkaline medium. In an alkaline medium, Fe³⁺ can induce a high-valency state in Ni, facilitating bonding with OH⁻ and promoting water dissociation during the OER. The selective adsorption of OH⁻ on Ni-sites forms OH-Ni^{δ+} intermediates, then decomposes to form O₂. The synergy between Ni and Fe species facilitates water dissociation for HER, with hydrogen adsorption on Ni sites and OH⁻ adsorption on Fe sites from H_{ad}-Ni^{δ+} and FeOOH intermediates. Additionally, the presence of Fe inhibits the self-oxidation of Ni and promotes the dissociation of water into H₂ during the HER process^{49,50}. The dispersion of MnO₂ nanorods on NiFe-LDH and SWCNT increases the electrochemically active sites and conductivity of the composite material, NiFe-LDH@3-MnO₂/SWCNT. This synergistic combination enhances the inherent OER and HER activity of NiFe-LDH@3-MnO₂/SWCNT^{51,52}.

A comparative analysis of the electrocatalytic performance of NiFe-LDH@3-MnO₂/SWCNT with other reported NiFe-LDH composites reveals its superior OER activity, as evident from Fig. 8a. MnO₂ played an important role in enhancing the OER performance of the NiFe-LDH. Since MnO₂ enables reversible transitions between the Mn²⁺, Mn³⁺, and Mn⁴⁺ oxidation states, it promotes OER. Additionally, the Mnⁿ⁺ increases the electron transfer from the electrolyte and forms OOH intermediates, which subsequently undergo deprotonation. Also, the presence of Mn³⁺ ions in MnO₂ enhances the adsorption of OH⁻ and accelerates the reaction intermediates. A comparative analysis of MnO₂-based composites for OER revealed that NiFe-LDH@3-MnO₂/SWCNT outperforms other materials, as evident from Fig. 8b. Notably, the Tafel slope of NiFe-LDH@3-MnO₂/SWCNT (33 mV/dec) is less than that of other MnO₂-based materials, including CoTe/MnO₂/BN (81 mV/dec)⁵³, Ni/MnO₂/CC (45 mV/dec)⁵⁴, and MnO₂ anchored N-Ti₃C₂ MXene (73.2 mV/dec)⁵⁵. This superior performance is attributed to the MnO₂ nanorods grown on conductive SWCNTs, which enhance the high-conductivity pathway, facilitating high electrolyte diffusion into the LDH layers and thereby reducing the liquid sealing effect. The hierarchical structure of NiFe-LDH@3-MnO₂/SWCNT composite further enhances the electrocatalytic surface area and high electrical conductivity. Although the HER activity of NiFe-LDH@3-MnO₂/SWCNT is lower when

Table 1 Comparison of the electrocatalytic activity of synthesized catalysts.

Catalysts	Overpotential@10 mA cm ⁻² (mV)		Tafel slope (mV dec ⁻¹)	
	HER	OER	HER	OER
NiFe-LDH	248	223	127	62
NiFe-LDH@1-MnO ₂ /SWCNT	225	211	121	46
NiFe-LDH@3-MnO ₂ /SWCNT	198	206	119	33
NiFe-LDH@5-MnO ₂ /SWCNT	216	221	121	41

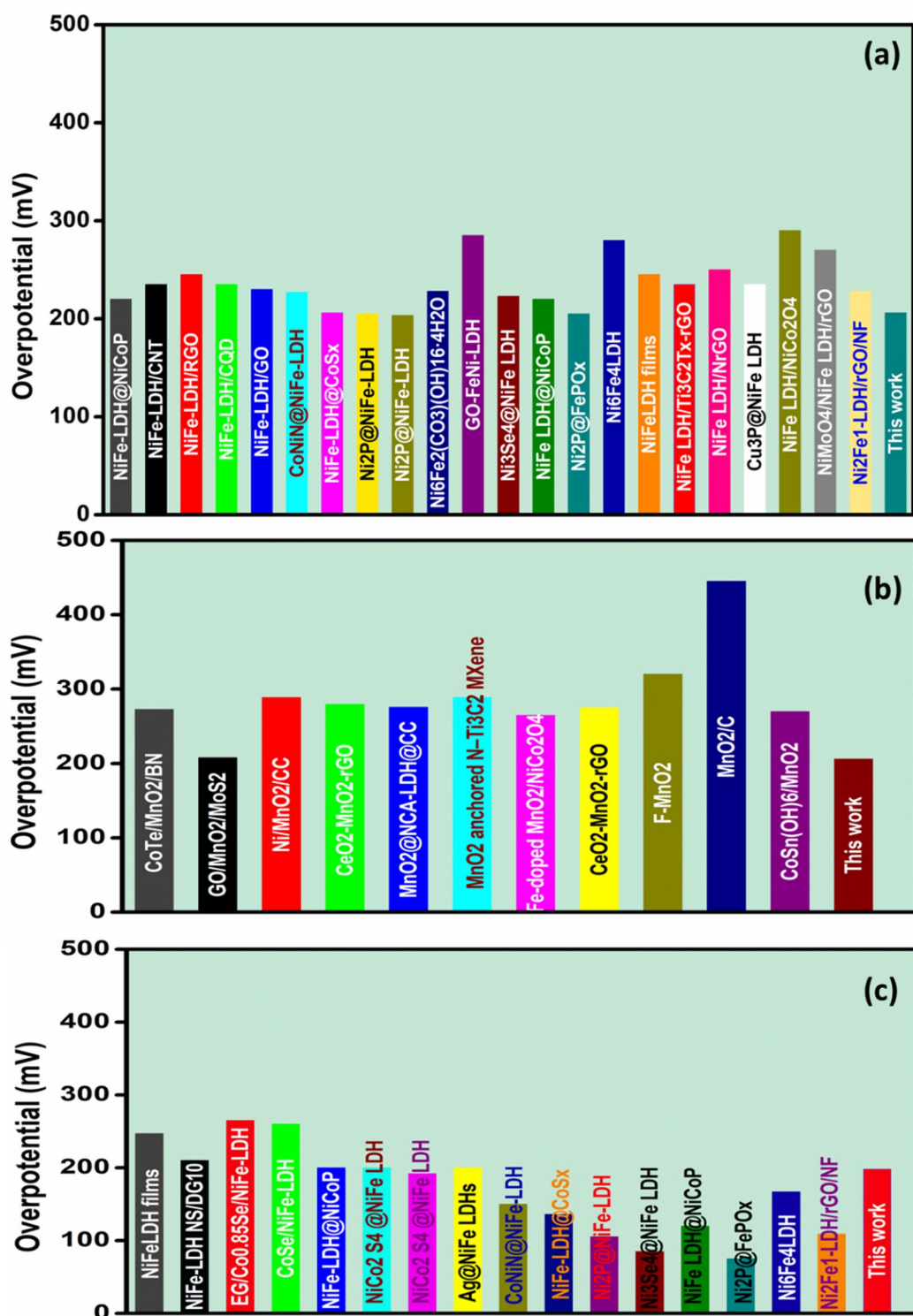


Fig. 8 (a) Comparative analysis of the electrocatalytic performance @ 10 mA of NiFe-LDH@3-MnO₂/SWCNT composite with previously reported NiFe-LDH composites for OER, (b) Comparison bar chart of composite's with prior reported MnO₂ composites for OER, and (c) Comparative analysis of the electrocatalytic performance @ 10 mA of NiFe-LDH@3-MnO₂/SWCNT composite with previously reported NiFe-LDH composites for HER.

compared to other catalysts, as shown in Fig. 8c, optimizing HER performance is important for overall water splitting. A slow HER process can increase the overpotential and decrease system efficiency. Future modifications, such as incorporating HER-active materials or surface engineering, can enhance performance. Overall, the use of earth-abundant transition metals, such as Ni, Fe, and Mn, reduces production costs, making large-scale applications more economically viable. With its improved electrocatalytic efficiency, stability, and durability, the NiFe-LDH@3-MnO₂/SWCNT is a promising candidate for electrochemical water splitting.

Conclusion

To summarize, this study successfully synthesized the hierarchical NiFe-LDH with intercalated CO₃²⁻ and its composites via a simple hydrothermal method. The morphological studies of the synthesized electrocatalysts were investigated using SEM and HR-TEM analysis. The results revealed that NiFe-LDH exhibited nanosheet and micro-flower morphologies. Furthermore, the in-situ growth of NiFe-LDH on MnO₂/SWCNT led to the uniform dispersion of MnO₂ nanorods and effective integration of SWCNT within the NiFe-LDH@3-MnO₂/SWCNT composite. Electrochemical studies revealed that the optimized NiFe-LDH@3-MnO₂/SWCNT exhibited low overpotentials of 206 and 198 mV at 10 mA cm⁻² for OER and HER, respectively. Additionally, the studies revealed that the heterojunction formed between NiFe-LDH, MnO₂, and SWCNT enhanced the conductivity and mass-transfer properties of NiFe-LDH@3-MnO₂/SWCNT. Notably, the NiFe-LDH@3-MnO₂/SWCNT showed enhanced OER performance compared to HER, which can be attributed to the role of MnO₂ in enhancing the OER performance of NiFe-LDH. MnO₂ facilitates a reversible transition among the Mn²⁺, Mn³⁺, and Mn⁴⁺ oxidation states, catalyzing the OER and achieving better performance than most reported NiFe-LDH-based electrocatalysts. Moreover, the fabricated two-electrode system retained approximately 98.63% of its initial performance after 48 h of continuous operation. The enhanced electrocatalytic activity is attributed to the hierarchical morphology and hetero-interfaces formed between NiFe-LDH, MnO₂, and SWCNT. Our work suggests that the NiFe-LDH@3-MnO₂/SWCNT can be a cost-effective and efficient catalyst for water-splitting applications.

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Data availability All data generated or analysed during this study are included in this published article.

Declarations

Competing interests The authors declare no competing interests.

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